

Predicates and Quantifiers

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4.1 Overview

Propositional logic cannot look inside a statement: from “all men are mortal” and “Socrates is a man” it cannot extract “Socrates is mortal”. Predicate logic repairs this by introducing *variables*, *predicates* and *quantifiers*, turning logic into a language rich enough for all of mathematics.

4.2 Predicates

Definition 4.1 (Predicate). A *predicate* is a statement $P(x_1, \dots, x_n)$ containing variables, which becomes a proposition once each variable is given a value from its *domain* (universe of discourse) U . A one-variable predicate is called a *propositional function* $P : U \rightarrow \mathbb{B}$.

Example 4.2. Over $U = \mathbb{Z}$, let $P(x)$: “ $x^2 \leq 9$ ”. Then $P(2)$ is true, $P(5)$ is false; $P(x)$ alone is neither — it is a function awaiting an argument.

Example 4.3 (Pythagorean triples). Let $T(a, b, c)$: “ $a^2 + b^2 = c^2$ ” over positive integers. Then $T(3, 4, 5)$ and $T(5, 12, 13)$ are true while $T(2, 3, 4)$ is false. The predicate packages infinitely many propositions into a single formula.

There are two ways to convert a predicate into a proposition: substitute concrete values, or *quantify* the variables.

4.3 The universal quantifier

Definition 4.4. The *universal quantification* $\forall x P(x)$ (“for all x , $P(x)$ ”) is true exactly when $P(a)$ is true for *every* element a of the domain. An element a with $P(a)$ false is a *counterexample* and makes $\forall x P(x)$ false.

Example 4.5. Over $U = \mathbb{R}$: $\forall x (x^2 \geq 0)$ is true. Over $U = \mathbb{Z}$: $\forall x (x^2 > 0)$ is false — the single counterexample $x = 0$ suffices.

When the domain is finite, $U = \{a_1, \dots, a_n\}$, the universal quantifier is a big conjunction: $\forall x P(x) \equiv P(a_1) \wedge \dots \wedge P(a_n)$.

4.4 The existential quantifier

Definition 4.6. The *existential quantification* $\exists x P(x)$ (“there exists an x such that $P(x)$ ”) is true exactly when $P(a)$ is true for *at least one* element a of the domain; a single *witness* suffices.

Example 4.7. Over $U = \mathbb{Z}$: $\exists x (x^2 = 4)$ is true (witness $x = 2$, or -2); $\exists x (x^2 = 3)$ is false.

Over a finite domain the existential quantifier is a big disjunction: $\exists x P(x) \equiv P(a_1) \vee \dots \vee P(a_n)$. The uniqueness variant $\exists! x P(x)$ asserts that exactly one witness exists.

The truth of a quantified statement depends on the domain. “ $\exists x (2x = 1)$ ” is false over $\mathbb{Z}$ but true over $\mathbb{Q}$. Always state the universe.

4.5 Quantifier equivalences

Theorem 4.8 (De Morgan laws for quantifiers).

$$\neg \forall x P(x) \equiv \exists x \neg P(x), \quad \neg \exists x P(x) \equiv \forall x \neg P(x).$$

Negation flips each quantifier as it moves inward. To negate a quantified statement: change every $\forall$ to $\exists$ and vice versa, then negate the matrix.

Theorem 4.9 (Distribution laws). $\forall$ distributes over $\wedge$, and $\exists$ over $\vee$:

$$\forall x (P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x), \quad \exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x).$$

Neither distributes over the other connective: $\forall x (P(x) \vee Q(x))$ is *not* equivalent to $\forall x P(x) \vee \forall x Q(x)$, and dually.

Example 4.10. Over $\mathbb{Z}$, with $P(x)$: “ x is even” and $Q(x)$: “ x is odd”: $\forall x (P(x) \vee Q(x))$ is true, but $\forall x P(x) \vee \forall x Q(x)$ is false.

Restricted quantification abbreviates implication and conjunction:

$$\forall x \in S, P(x) \text{ means } \forall x (x \in S \rightarrow P(x)), \quad \exists x \in S, P(x) \text{ means } \exists x (x \in S \wedge P(x)).$$

4.6 Nested quantifiers

Quantifiers may be stacked; **order matters** when the quantifiers differ.

Example 4.11. Over $\mathbb{R}$:

$$\forall x \exists y (x + y = 0)$$

is true ($y = -x$ depends on x), while

$$\exists y \forall x (x + y = 0)$$

is false (no single y works for all x). Same-type quantifiers commute: $\forall x \forall y \equiv \forall y \forall x$.

Example 4.12 (The limit definition). “The sequence (a_n) converges to L ” is the triply-quantified statement

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N, |a_n - L| < \varepsilon,$$

whose negation, by repeated use of Theorem 4.8, is $\exists \varepsilon > 0 \forall N \exists n \geq N, |a_n - L| \geq \varepsilon$. Fluency with nested quantifiers is exactly what analysis requires (Chapter 9).

4.7 Translating statements

Example 4.13. Domain: all people. $S(x)$: “ x is a student”; $C(x)$: “ x has a computer”; $F(x, y)$: “ x and y are friends”.

- “Every student has a computer”: $\forall x (S(x) \rightarrow C(x))$.
- “Some student has no computer”: $\exists x (S(x) \wedge \neg C(x))$.
- “Everybody has a friend”: $\forall x \exists y F(x, y)$.
- “There is someone who is friends with everyone”: $\exists x \forall y F(x, y)$.
- “No two distinct people have exactly the same friends”: $\forall x \forall y (x \neq y \rightarrow \exists z (F(x, z) \oplus F(y, z)))$.

Pattern to memorise: universal statements pair with $\rightarrow$ (“all S are C ”: $\forall x(S(x) \rightarrow C(x))$), while existential statements pair with $\wedge$ (“some S is C ”: $\exists x(S(x) \wedge C(x))$). Mixing these up is the most common translation error.

4.8 Summary

Predicates are truth-valued functions of variables; quantifiers $\forall$ and $\exists$ bind those variables to produce propositions. Negation exchanges the two quantifiers (De Morgan); $\forall$ distributes over $\wedge$, $\exists$ over $\vee$; the order of unlike nested quantifiers changes meaning. Universal claims are refuted by one counterexample; existential claims are proved by one witness.

Exercises

Exercise 4.1 EASY

Let $P(x)$: “ $x^2 = x$ ” over the domain $\mathbb{Z}$. Determine the truth values of $P(0)$, $P(1)$, $P(2)$, $\exists x P(x)$ and $\forall x P(x)$.

Exercise 4.2 EASY

Write the negation of each statement, with quantifiers pushed inward: (a) $\forall x (x^2 > x)$; (b) $\exists x (x^2 = 2)$.

Exercise 4.3 EASY

Over the domain of all animals, with $C(x)$: “ x is a cat” and $M(x)$: “ x chases mice”, translate: (a) “every cat chases mice”; (b) “some cat does not chase mice”. Then negate (a) symbolically and in English.

Exercise 4.4 MEDIUM

Determine, with justification, the truth value over $\mathbb{Z}$ of: (a) $\forall n \exists m (m > n)$; (b) $\exists m \forall n (m > n)$; (c) $\forall n \exists m (n + m = 0)$; (d) $\exists m \forall n (nm = n)$.

Exercise 4.5 MEDIUM

Show by a concrete domain and predicates that $\forall x (P(x) \vee Q(x))$ is not equivalent to $\forall x P(x) \vee \forall x Q(x)$, and that $\exists x (P(x) \wedge Q(x))$ is not equivalent to $\exists x P(x) \wedge \exists x Q(x)$.

Exercise 4.6 MEDIUM

Express using quantifiers over $\mathbb{Z}$: (a) “ n is even”; (b) “ n is a perfect square”; (c) “ n is prime” (you may use $|$ for divisibility); (d) “there is no largest integer”.

Exercise 4.7 MEDIUM

Write the negation of the convergence statement $\forall \varepsilon > 0 \exists N \forall n \geq N, |a_n - L| < \varepsilon$, justifying each quantifier flip.

Exercise 4.8 HARD

Let $F(x, y)$: “ x trusts y ” over a set of people. Express: (a) “everyone trusts somebody, but nobody trusts everybody”; (b) “there are two distinct people who trust each other”; (c) “trust is never mutual”. Can (b) and (c) both be true? Explain.

Exercise 4.9 HARD

Prove that $\exists x \forall y P(x, y) \models \forall y \exists x P(x, y)$, and give a counterexample showing the converse entailment fails.

Solutions to the Exercises

Solution 4.1. $x^2 = x$ holds iff $x(x - 1) = 0$ iff $x \in \{0, 1\}$. So $P(0)$, $P(1)$ are true, $P(2)$ is false; $\exists x P(x)$ is true (witness 0); $\forall x P(x)$ is false (counterexample 2). ■

Solution 4.2. (a) $\exists x (x^2 \leq x)$. (b) $\forall x (x^2 \neq 2)$. ■

Solution 4.3. (a) $\forall x (C(x) \rightarrow M(x))$. (b) $\exists x (C(x) \wedge \neg M(x))$. Negation of (a): $\neg \forall x (C(x) \rightarrow M(x)) \equiv \exists x (C(x) \wedge \neg M(x))$ — “some cat does not chase mice”, which is exactly (b). ■

Solution 4.4. (a) True: given n , take $m = n + 1$. (b) False: for any m , $n = m$ fails $m > n$. (c) True: take $m = -n$. (d) True: $m = 1$ satisfies $n \cdot 1 = n$ for all n . ■

Solution 4.5. Over $\mathbb{Z}$ with $P = \text{"even"}$, $Q = \text{"odd"}$: every integer is even or odd, so $\forall x(P \vee Q)$ is true, while neither $\forall xP$ nor $\forall xQ$ holds. For the second pair: $\exists xP(x)$ and $\exists xQ(x)$ are both true, but no integer is both even and odd, so $\exists x(P \wedge Q)$ is false. ■

Solution 4.6. (a) $\exists k(n = 2k)$. (b) $\exists k(n = k^2)$. (c) $n > 1 \wedge \forall d(d \mid n \rightarrow (d = 1 \vee d = n))$. (d) $\neg \exists m \forall n(n \leq m)$, equivalently $\forall m \exists n(n > m)$. ■

Solution 4.7. Negate step by step: $\neg \forall \varepsilon \Phi \equiv \exists \varepsilon \neg \Phi$; $\neg \exists N \Psi \equiv \forall N \neg \Psi$; $\neg \forall n \geq N \chi \equiv \exists n \geq N \neg \chi$; finally $\neg(|a_n - L| < \varepsilon) \equiv |a_n - L| \geq \varepsilon$. Result:

$$\exists \varepsilon > 0 \forall N \exists n \geq N, \quad |a_n - L| \geq \varepsilon.$$

■

Solution 4.8. (a) $\forall x \exists y F(x, y) \wedge \neg \exists x \forall y F(x, y)$. (b) $\exists x \exists y (x \neq y \wedge F(x, y) \wedge F(y, x))$. (c) $\forall x \forall y (x \neq y \rightarrow \neg(F(x, y) \wedge F(y, x)))$. (b) and (c) are direct negations of each other (over pairs of distinct people), so they cannot both be true in the same situation (assuming at least two people). ■

Solution 4.9. Entailment: suppose $\exists x \forall y P(x, y)$, with witness x_0 . Given any y , $P(x_0, y)$ holds, so $\exists x P(x, y)$ holds; as y was arbitrary, $\forall y \exists x P(x, y)$. Converse fails: over $\mathbb{R}$ with $P(x, y)$: “ $x + y = 0$ ”, $\forall y \exists x$ is true ($x = -y$) but no single x works for every y , so $\exists x \forall y$ is false. ■